INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 6: Conclusion and Future Work

by

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6 Conclusion and Future Work

Chapter 5 interpreted the findings and judged whether the study met its objectives.

6.1 Answers to the research questions

The thesis proposed EndorseRank, PageRank with the edge weights of Adaptive Weighted PageRank (AWP) on a latest-allowance endorsement graph, and evaluated it beside AWP itself (T. Do et al., 2023), a coupled walk over both edge types and the raw degrees these walks smooth, on Arbitrum One contracts from October 2023 to June 2026, with holdout windows W0 (April to June 2026) and W1 (July to September 2026). Amounts were revised to US dollars on five tokens, and the cohort with them, after the registered results were known; the rule verdicts below are a re-analysis matching the registered ones (Appendix 8.6).

RQ1 asked whether EndorseRank and AWP rank the contracts of one sample differently. They do, and H1 is supported, largely by construction (Section 4.4). Whether the two reflect different kinds of reputation rests mainly on the holdouts of RQ4.

RQ2 asked how closely each score agrees with allowance and transfer statistics in the same window. Each score follows the edges it walks on (Table 4.3), which supports H2 only narrowly, because the proxies come from the same events as the graphs.

RQ3 asked what each score costs to compute. The cost grows with the size of each graph, so EndorseRank, on the sparsest graph, is the cheapest score, and the coupled walks cost more than AWP (Table 4.1); H3 is supported, largely by construction. On subsamples the time ratio follows the edge ratio only at the two largest stages (Table 4.2).

RQ4 asked whether scores frozen at t_1 predict the new approval pairs and new transfer senders that spender contracts gain afterwards, and whether any PageRank predicts them better than the raw

degree it smooths. The first part holds in both windows, and over all spenders the edge type decides which label a score anticipates (Tables 4.10 and 4.14). The second differs by walk, in contrasts computed after the labels were known: AWP ranks later new senders above the transfer in-degree in both windows, and EndorseRank ranks later approval pairs below the in-approve degree in both (Tables 4.11 and 4.16). With AWP's activity restarts EndorseRank ranked later approval pairs above the degree in both windows, a post hoc hypothesis, not a finding.

RQ5 asked whether a coupled PageRank over both edge types improves on a walk over the transfers alone, and whether it ranks contracts better than either layer walked alone. For C-PR, the coupling the plan fixed, both answers are no. Under rule A, C-PR meets the allowance condition of the primary part but fails its transfer condition and the secondary part (Section 4.3.1; Table 4.11). Under rule B, its exploratory gain over the transfer walk on new approval pairs in W0 recurred in W1, possibly in part through the spenders that walk scores zero, but its loss on new senders exceeded the margin, and on the traders' liquidation-free close rate a loss beyond it could not be ruled out; against AWP it ranked new approval pairs higher and new senders lower, with that close rate unresolved (Table 4.15). Post hoc, that lead comes from the spenders AWP scores zero and most of the loss from C-PR's transfer layer, not the coupling, while C-AWP, coupling AWP with EndorseRank under activity restarts, leads AWP on approval pairs among the others too, yet fails rule B's structure (Section 4.6.4). The seeded ablation matches the transfer walk on new senders: in this form, allowance edges contribute mainly when the walk moves along them, and little as a seed.

In a comparison registered with rule B that adds no research question (Section 5.1.6), the approval count at the freeze led the transfer count on later liquidation avoidance in W0, but in W1, the primary window, the interval of the lead includes zero ($P = +0.084$, 97.5% interval $[-0.026, 0.188]$; Section 4.6.3). The like-for-like PageRank pair is not resolved, and the approval count is lower on the profitable share of closes, so its signal concerns forced-liquidation avoidance, not trading success. The samples are small.

The main research question asked what ERC-20 allowance edges add to PageRank as a reputation score for contract accounts on Arbitrum One, compared with the transfer walk and with the raw

degrees that both walks smooth. Against the transfer walk they add a second construct, delegated authority, which orders the contracts differently and reaches the one spender in six that transfer walks score zero. In comparisons computed after the labels were known, their lead over AWP on later approval pairs rests on those spenders (Table 4.12), and the allowance walk with uniform restarts ranked those pairs below the in-approve degree in both windows.

6.2 Original contribution to knowledge

Read against the literature of Chapter 2, the five contributions of Section 1.9 establish the following. Before this work, graph reputation on public ledgers ranked wallets by their transfers (H.-D. Do & Do, 2023; T. Do et al., 2023), by how risk spreads through transaction networks (D. Lin et al., 2025), or by explicit ratings (Almasoud et al., 2020; Bellini et al., 2020), and the reputation tooling deployed in DeFi scored the holder of an address from credentials (Human Passport, n.d.). None of it used the ERC-20 allowance, the on-chain record of a deliberate delegation of spending power (Vogelsteller & Buterin, 2015), as the edge of a reputation ranking, or ranked contract accounts by the approvals they receive; approvals had been used to cluster the addresses of one entity (Victor, 2020) and studied as a security risk (Wang et al., 2022). None evaluated on-chain reputation scores on one cohort against the checks of each construct and against the raw degrees they smooth, with confidence intervals, difference tests and temporal holdouts under decision rules fixed in advance. This thesis supplies both; PageRank and AWP's weighting are taken from the literature (T. Do et al., 2023; Page et al., 1999), EndorseRank changes only the edge, and the coupled operator changes the walk.

The new knowledge has five parts. What an on-chain reputation score of a contract measures depends on the edges it uses, and this can be checked on one cohort with quantified uncertainty. The allowance edge yields a score that differs from the transfer-based score, ranks later endorsements in two windows and orders the spender contracts without transfer history, which transfer walks score zero; its post hoc lead over AWP there rests on them (Table 4.12; Section 4.6.4). On few traders, the

comparison registered with rule B found, in the re-analysis as in the registered analysis, no count-level lead of allowance over transfer on later avoidance of forced liquidation in the primary window and left the PageRank pair unresolved. Two further steps that the literature suggested did not consistently improve results on these data: in contrasts computed after the labels were known, propagation beat the raw degree it smooths out of window with activity restarts but not with uniform restarts, on either edge type, so the restart rule, not the edge type, appears to decide whether it does; the fixed-λ coupling of the plan beats neither its better single layer on both spender labels nor, under rule B, the transfer-only walk. Finally, the allowance-side result has a partial analytic explanation: at depth one EndorseRank is, up to a constant, a linear function of the owner-normalized in-strength (Section 5.3.2). The construct-validity framing (Cronbach & Meehl, 1955; Messick, 1995) that makes these claims testable is part of the contribution, as is the open pipeline whose published summaries let others check every table, since whether decentralized systems miscalculate credit scores is a matter of public interest (Packin & Lev-Aretz, 2024).

The thesis does not show that on-chain reputation can replace collateral (Gudgeon, Perez, et al., 2020; Schär, 2021), that the results apply beyond contracts approved on the five tokens on Arbitrum One in the window studied, or that any score measures creditworthiness, default risk or trading skill, and it does not claim that EndorseRank resists Sybils: under uniform teleportation a closed farm of attacker addresses can inflate it cheaply (Section 5.6).

6.3 Future work

Each suggestion below answers a limitation of Chapter 5. Any new score or weighting it leads to should be fixed before it is scored, replicated on registered new data and checked out of window against degree baselines.

- A registered test of the revised analysis. A registration committed before the logs of a later window, such as October to December 2026, are extracted would fix its US-dollar amounts, five tokens, value slopes and allowance cap with the cohort rule, applied only to events up

to its freeze, and with rules A and B and the comparison on traders, which needs more contract traders than W0 and W1 held. The value-slope check did not cover the holdout labels (Table 4.7), so the slopes and the cap should not be tuned on the new window.

- Propagation on a deeper sample. On the ego networks studied, EndorseRank reduces to a reweighted approval count for every depth-one spender (Section 5.3.2); a sample extracted two or more hops out would test whether any propagated score adds out-of-window value over the local count, on arrival labels and on labels that are not counts of arrivals. The variant with AWP's activity restarts should be tested against the in-approve degree on new data, separately for depth-one and deeper spenders, since only deeper spenders can show whether propagation, and not the restarts alone, adds to the count.
- Validate the results in other contexts. Running the same analysis on other rollups, on Layer-1 networks and over later windows would show whether the results extend beyond Arbitrum One and the months studied; because DeFi is heterogeneous (Schär, 2021; Werner et al., 2022), this is the main threat to external validity. Scoring externally owned accounts lies outside this thesis.
- Adversarial evaluation. The Sybil cost model is analytical, and no attack was carried out on chain (Section 5.6). Restricting teleportation and dangling redistribution to a vetted seed set, as TrustRank does (Gyöngyi et al., 2004), would remove the uniform inflow that closed farms exploit and should be evaluated under the same attacks. Red-teaming with wash approvals, spender farms and strategic revocation would measure what attacks cost under live fee markets and whether the Sybil-adjusted proxies stay informative.
- Coupling beyond a fixed transition-level mixture. Besides C-PR, only C-AWP was tried, after the labels were known (Section 4.6.4); it is the coupling to register on a later window such as October to December 2026. Node-dependent or learned weights, per-layer damping and mixtures of the two stationary vectors remain untested; any such design should report construct-specific checks and keep the auditability that sets graph propagation apart from

opaque learned scores (Kotb, 2023; Packin & Lev-Aretz, 2024).

- Composition with attestation tooling. Filtering the approving owners through a Passport-style gate (Human Passport, n.d.), which answers whether an address belongs to a distinct person, before EndorseRank orders the contracts by the authority those owners delegate is a plausible deployment; it should be evaluated on labeled Sybil data, with the gate's own weaknesses (Bordeianu & Popescu, 2026; Siddarth et al., 2020) in view.
- Deployment studies. Integrating EndorseRank with risk oracles and governance tooling would test whether its runtime profile can sustain production refresh cadences. Controlled experiments on fee tiers or monitoring cadence would let evidence replace the adoption hypotheses of Section 5.4.4.

6.4 Closing reflection

In adversarial, pseudonymous markets on-chain reputation does not remove the need for collateral (Gudgeon, Perez, et al., 2020), but it can make behavioral structure readable, and which structure it reads depends on the edges used. A transfer records economic flow and an allowance records delegated authority. PageRank will rank either graph, yet one ranking does not serve both constructs, and C-PR's walk over both graphs gained little over the right single graph on one label and lost to it on the other; a coupling that keeps AWP's weights and restarts did better post hoc and is the one to test on a later window. For a protocol or researcher who needs a fast, auditable signal of which contracts owners have authorized, the allowance edge is the one to use on these data, above all for spenders without a transfer edge. The approval count is its cheapest form and, post hoc, ranked later approvals better than EndorseRank with uniform restarts, and AWP better still among the W0 spenders that had received a transfer; EndorseRank is the propagated form for those who want the endorsers to count as well, with its exposure to cheap inflation by closed farms in mind (Section 5.6). For describing transfer hubs and anticipating their inflow, the transfer edge is the one to use, and there, also post hoc, AWP with its activity restarts improved on the count. For screening

counterparties against forced liquidation, the approval count led the transfer count in W0 but not resolvably in the primary window, and settling it needs more contract traders.